Daily Research Summary: GU-RSVP Cosmological Fitting Pipeline

[Redacted]

June 12, 2025

Abstract

This summary outlines progress on the Geometric Unity (GU) and Relativistic Scalar Vector Plenum (RSVP) synthesis, transitioning from a simulation framework to a cosmological data-fitting pipeline. The pipeline operationalizes Weinstein's critique of General Relativity's static formalism by implementing a computable alternative to ΛCDM, grounded in RSVP's entropic dynamics. Key deliverables include a Python-based pipeline integrating $\Lambda_{\text{eff}}(z)$ extraction, redshift mapping, and Cobaya fitting against DESI BAO (2024) and Pantheon+ SNIa datasets, with static visualization outputs.

1 Summary

Today's session finalized a cosmological fitting pipeline for the GU-RSVP framework, enabling falsifiable tests of its predictions. The pipeline includes:

- Numerical extraction of $\Lambda_{\text{eff}}(t) \sim \partial_t(\nabla \cdot \vec{v}) - \partial_t S$ from 3+1D RSVP simulation traces (S, v).
- Mapping of simulation time t to redshift z via $z(t) = \exp\left(\int H(t')\,dt'\right) - 1$, with $H(t) \approx \nabla \cdot \vec{v}/3$.
- Spline interpolation of $\Lambda_{\text{eff}}(z)$ for flexible parametrization.
- Cobaya-based MCMC fitting against DESI BAO (2024) and Pantheon+ SNIa datasets.
- Posterior sampling for Λ_0, γ, H_0, Ω_m, and derived $w(z)$.

The pipeline supports tests of deviations from a constant Λ, consistency with DESI's $w_0 > -1$, $w_a < 0$, and potential reconciliation of the Hubble tension.

2 Framework Alignment

- **RSVP Dynamics**: Scalar (Φ), vector ($\vec{v}$), and entropy (S) fields evolve via local updates, with Λ_{eff} driven by $\nabla \cdot \vec{v}$ and S gradients.
- **Emergent Geometry**: Metric structure encoded in $\mathcal{R}_{\text{eff}} \sim R + \lambda_1(\delta S/\delta\Phi)^2 + \lambda_2 v^A v^B R_{AB}$.
- **GU Symmetries**: TARTAN lattice and $\vec{v}$ degeneracy intended to yield Standard Model gauge groups (future implementation).
- **Testability**: $\Lambda_{\text{eff}}(z)$ compared directly to ΛCDM using cosmological data.

3 Next Steps

1. **Empirical Calibration**: Integrate Planck CMB lensing and SH0ES H_0 priors to address the Hubble tension.
2. **Tensor Extension**: Simulate full field equations with curvature tensors R_{AB}.
3. **Visualization**: Develop interactive 3D/4D viewers for $\vec{v}$ streamlines and S fields.
4. **Phenomenological Interpretation**: Relate $\Lambda_{\text{eff}}(z)$ to structure formation, lensing, or B-mode polarization.

4 Conclusion

The GU-RSVP framework now supports a testable cosmological pipeline, realizing Weinstein's vision of a dynamic substrate. By fitting $\Lambda_{\text{eff}}(z)$ to high-precision data, it addresses foundational issues in spacetime geometry and engages with empirical challenges like the Hubble tension and evolving dark energy, moving toward falsifiability and scientific utility.